Auteur: X Tenka Tenzin Choezin
Studierichting: Informatica
Oefeningenreeks 1B nr. 44.2

$z^3 - 5(1+2i)^2 + 24 - 10i = 0$

Stel $t = z^2 \Rightarrow t^2 - (5+10i)t + 24 - 10i = 0$

De nulpunten zoeken

$D = (-5-10i)^2 - 4 \cdot 1 \cdot (24 - 10i)$

$= 25 + 100i - 100 - 96 + 40i$

$= -171 + 140i$

Discriminant is een complex getal.

$$-171 + 140i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = -171 \\ 2xy = 140 \Rightarrow x^2(-y^2) = 4900 \end{cases}$$

$\lambda^2 + 171\lambda - 4900 = 0$

Berekening van nulpunten:

$D = 171^2 - 4 \cdot 1 \cdot (-4900) = 48841 \Rightarrow \sqrt{100} = 221$

$$\lambda_{1,2} = \frac{-171 \pm 221}{2}$$

$\lambda_1 = 25$ en $\lambda_2 = -196$

$x = \pm 5$ en $y = \pm 14i$

$xy$ is positief $\Rightarrow$ vierkantswortel van eerste discriminant: $5 + 14i$ of $-5 - 14i$

$$t_{1,2} = \frac{5 + 10 \pm (5 + 14i)}{2}$$

$t_1 = 5 + 12i$ en $t_2 = -2i$

$t = z^2 \Leftrightarrow z = \sqrt{t}$ :

**Voor** $t = 5 + 12i$

$5 + 12i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = 5 \\ 2xy = 12 \Rightarrow x^2(-y^2) = 36 \end{cases}$

$\lambda^2 - 5\lambda - 36 = 0$

Berekening van nulpunten:

$D = 25 - 4 \cdot 1 \cdot (-36) = 169 \Rightarrow \sqrt{169} = 13$

$\lambda_{1,2} = \dfrac{5 \pm 13}{2}$

$\lambda_1 = 9$ en $\lambda_2 = -4$

$x = \pm 3$ en $y = \pm 2$

$xy$ is positief $\Rightarrow z_1 = 3 + 2i$ en $z_2 = -3 - 2i$

**Voor** $t = -2i$

$-2i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \Rightarrow x^2(-y^2) = 1 \end{cases}$

$\lambda^2 - 1 = 0$

$\lambda^2 = 1$

$\lambda_{1,2} = \pm 1$

$x = \pm 1$ en $y = \pm 1$

$xy$ is negatief $\Rightarrow z_3 = 1 - i$ en $z_4 = -1 + i$

# References